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# Shelf algorithms for on-line strip packing [1]

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**Abstract**

In the strip packing problem, the goal is to pack a set of rectangles into a vertical strip of unit width so as to minimize the total height of the strip needed. For the on-line version of this problem, Baker and Schwarz introduced the class of so-called shelf algorithms. One of these shelf algorithms, FFS, is the current champion for on-line strip packing. The asymptotic worst case ratio of FFS can be made arbitrarily close to 1.7. We show that no shelf algorithm for on-line strip packing can have an asymptotic worst case ratio better than $h_\infty \approx 1.69103$. Moreover, we introduce and analyze another on-line shelf algorithm whose asymptotic worst case ratio comes arbitrarily close to $h_\infty$. © 1997 Elsevier Science B.V.

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## 1. Introduction

**Problem statement.** In the *strip packing* problem, the input consists of a list of rectangles $r_i = (h_i, w_i)$, $1 \leqslant i \leqslant n$, with height $h_i$ and width $w_i$. Height and width of every rectangle are values in $[0, 1]$. The goal is to pack this list into a vertical strip of unit width so as to minimize the total height of the strip needed. The rectangles must be packed orthogonally with their sides parallel to the axes; rotations are forbidden. The strip packing problem arises in a wide variety of contexts and has many applications e.g. in cutting stock problems and in VLSI design. Since the problem of finding an optimal packing is NP-hard, research has concentrated on approximation algorithms that find near-optimal packings.

**Approximation algorithms and on-line algorithms.** Let $\mathrm{OPT}(L)$ and $A(L)$ denote, respectively, the height of the strip used by an optimum algorithm and the height of the strip used by an approximation algorithm $A$ to pack the input list $L$. Then the *absolute worst case ratio* $R_A$ of algorithm $A$ is defined as

$$R_A = \sup_L \{A(L)/\mathrm{OPT}(L)\}, \tag{1}$$

and the *asymptotic worst case ratio* $R_A^\infty$ of algorithm $A$ is defined as

$$R_A^\infty = \lim_{k\to\infty} \sup_L \{A(L)/\mathrm{OPT}(L) \mid \mathrm{OPT}(L) \geqslant k\} \tag{2}$$

(cf. [5]). These ratios are customarily used to measure the quality of approximation algorithms for packing problems. It turned out that for strip packing, the

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asymptotic worst case ratio $R_A^\infty$ is the more appropriate measure: It is robust against anomalies with a small number of rectangles in the optimum packing, and while packing the first few rectangles it allows the packing algorithm more freedom than the absolute worst case ratio does. Therefore, we will only deal with the asymptotic worst case ratio throughout this note.

Now assume an environment where the list $L$ of rectangles arrives one by one. When rectangle $r_i$ arrives, it must immediately and irrevocably be assigned to its place in the strip, and the next rectangle $r_{i+1}$ becomes only known *after* rectangle $r_i$ has been assigned. Such an environment is called *on-line*, and an approximation algorithm that is able to work in an on-line environment is called an *on-line* algorithm. In contrast to this type of algorithm are the *off-line* algorithms that may assume knowledge of the entire problem instance before producing any output.

**Known results.** Strip packing was first studied in 1980 by Baker, Coffman and Rivest [3] who derived the first off-line approximation algorithms for this problem. In 1981, Baker, Brown and Katseff [1] gave an off-line algorithm with asymptotic worst case ratio 5/4, and this algorithm remained the off-line champion for strip packing until 1996 when Kenyon and Remila [7] designed a fully polynomial time approximation scheme for this problem (with respect to the asymptotic worst case ratio). For the on-line version, Baker and Schwarz [4] introduced the so-called *shelf algorithms* (see Section 2 for a precise definition). One of these shelf algorithms, the FFS with asymptotic worst case ratio arbitrarily close to 1.7, is the current champion for on-line strip packing. Baker, Brown and Katseff [2] investigated lower bounds on the asymptotic worst case ratio of on-line packing algorithms. Combining their insights with a result of Van Vliet [10] on one-dimensional bin packing yields that no on-line strip packing algorithm can have an asymptotic worst case ratio better than 1.5401.

**Our results and organization of the paper.** Section 2 gives some definitions and introduces the concept of shelf algorithms. In Section 3, we show that $h_\infty \approx 1.69103$ is a lower bound on the asymptotic worst case ratio of *any* shelf algorithm (compare this bound to the 1.7 of the FFS algorithm!). In Section 4, we design and analyze a shelf algorithm whose asymptotic worst case ratio can be made arbitrarily close to the bound $h_\infty$. Section 5 closes with a short discussion.

## 2. Definitions

In a strip packing of a set of rectangles, a *shelf* is a rectangular part of the strip with width one and height at most one such that (i) every box is either completely inside or completely outside of the shelf and such that (ii) every vertical line through the shelf intersects at most one rectangle. Hence, the situation looks very much like a couple of books sitting on a book shelf. The basic idea of the *shelf algorithms* [4] is to classify the arriving rectangles according to their heights. The packing is then constructed as a sequence of shelves, and rectangles of similar heights are packed into the same shelves.

To be more precise, every shelf algorithm is based on an on-line approximation algorithm $A$ for one-dimensional bin packing (one-dimensional bin packing: given a list of items $a_1, \ldots, a_n \in [0,1]$, find a packing of these items into a minimum number of unit capacity bins) and on a real parameter $\alpha$, $0 < \alpha < 1$. Such a shelf algorithm is abbreviated by $\text{SHELF}(A, \alpha)$, and the one-dimensional algorithm $A$ is called its *slave* algorithm. Whenever a new rectangle $(h_i, w_i)$ arrives, the shelf algorithm determines the integer $s$ for which $\alpha^{s-1} < h_i \leqslant \alpha^s$ holds and puts the rectangle into the class $C_s$. All rectangles in $C_s$ are packed into shelves of height $\alpha^s$. Since the total width of all rectangles in a common shelf must be bounded by 1, the decision on which rectangles are packed together into a common shelf is exactly a one-dimensional bin packing problem. For every class $C_s$, this one-dimensional problem is solved by the slave algorithm $A$.

For example, the FIRST FIT SHELF (FFS) algorithm of Baker and Schwarz packs every class $C_s$ according to the FIRST FIT (FF) algorithm (cf. [6]): A new rectangle in $C_s$ is packed into the lowermost shelf of height $\alpha^s$ into which it will fit. In case no shelf of height $\alpha^s$ has sufficient space, a new shelf of height $\alpha^s$ is opened on top of the current packing. Hence, in our notation FFS becomes $\text{SHELF}(\text{FF}, \alpha)$.

Next, we define a sequence $t_i$ of integers that is well-known in on-line bin packing and that will also

turn out to be useful for our paper.

$$t_1 = 2,$$
$$t_{i+1} = t_i(t_i - 1) + 1 \quad \text{for } i \geqslant 1.$$

This sequence starts with $\langle 2, 3, 7, 43, 1807, 3263443, \ldots \rangle$ and its growth is doubly exponential. With this, we can finally give an exact definition of the number $h_\infty$.

$$\begin{aligned} h_\infty &= \sum_{i=1}^{\infty} \frac{1}{t_i - 1} \\ &= 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{42} + \frac{1}{1806} + \cdots \\ &\approx 1.69103. \end{aligned} \tag{3}$$

## 3. A lower bound for all shelf algorithms

In this section, we show that the shelf algorithms of Baker and Schwarz [4] with worst case ratio 1.7 are not far away from the best shelf algorithms: No shelf algorithm can have a worst case ratio better than $h_\infty$. Note that our result also holds for *off-line* shelf algorithms.

**Theorem 1.** *For any real $\alpha$, $0 < \alpha < 1$, and for any on-line algorithm $A$ for one-dimensional bin packing, the shelf algorithm $S = \text{SHELF}(A, \alpha)$ has asymptotic worst case ratio $R_S^\infty \geqslant h_\infty$.*

**Proof.** Suppose, there exists a real number $\alpha$, $0 < \alpha < 1$, and an on-line algorithm $A$ such that the corresponding shelf algorithm $S = \text{SHELF}(A, \alpha)$ has asymptotic worst case ratio $R_S^\infty = h_\infty - 2\delta$ where $\delta > 0$. Then there exists a sufficiently large threshold $H$ such that

$$S(L)/\text{OPT}(L) \leqslant h_\infty - \delta \quad \text{whenever } \text{OPT}(L) \geqslant H \tag{4}$$

holds, where $L$ is a finite list over $[0, 1]^2$. We determine the smallest index $d$ that fulfills the inequality

$$h_\infty - \delta \leqslant \sum_{i=1}^{d} \frac{1}{t_i - 1}, \tag{5}$$

and we define the rational number $\varepsilon$ as

$$\varepsilon = \frac{1}{d}\left(1 - \sum_{i=1}^{d} \frac{1}{t_i}\right). \tag{6}$$

A straightforward calculation shows that

$$\varepsilon = \frac{1}{t_d(t_d - 1)d}. \tag{7}$$

Finally, we define positive integers $b_i$ and $c_i$ for $1 \leqslant i \leqslant d$, such that

$$\alpha^i < \frac{b_i}{c_i} < \alpha^{i-1}, \quad 1 \leqslant i \leqslant d. \tag{8}$$

Let $B = \prod_{i=1}^{d} b_i(t_i - 1)$ and let $n_i = HBc_i/b_i$, for $1 \leqslant i \leqslant d$. Note that by the definition of $B$, every $n_i$ is integer and divisible by $(t_i - 1)$. We now define homogeneous lists $L_i$, $1 \leqslant i \leqslant d$, that consist of $n_i$ identical items as follows:

$$L_i = \left\langle \left(\frac{1}{t_i} + \varepsilon, \frac{b_i}{c_i}\right), \left(\frac{1}{t_i} + \varepsilon, \frac{b_i}{c_i}\right), \ldots, \left(\frac{1}{t_i} + \varepsilon, \frac{b_i}{c_i}\right) \right\rangle, \quad |L_i| = n_i. \tag{9}$$

The list $L = L_1 L_2 \ldots L_d$ is the concatenation of these $d$ lists. It is easy to see that

$$\text{OPT}(L) = HB \tag{10}$$

holds: Stacking all items of list $L_i$ one above the other yields a column of height $n_i b_i / c_i = HB$ and of width $1/t_i + \varepsilon$. The total width of all these $d$ columns equals $d\varepsilon + \sum_{i=1}^{d} 1/t_i$ which by Eq. (6) is equal to 1. Hence, all $d$ columns can be packed side by side into a strip of width 1 and height $HB$.

Because of (8), algorithm $S$ uses shelves of height $\alpha^{i-1}$ for packing the items in list $L_i$, and thus every list $L_i$ has its own corresponding shelf type. Since for $1 \leqslant i \leqslant d$,

$$\begin{aligned} &(t_i - 1)\left(\frac{1}{t_i} + \varepsilon\right) \\ &\quad = 1 - \frac{1}{t_i} + (t_i - 1)\frac{1}{t_d(t_d - 1)d} < 1 \end{aligned} \tag{11}$$

holds, there fit exactly $t_i - 1$ items of $L_i$ into a shelf of width 1 and height $\alpha^{i-1}$. Therefore,

$$S(L) = \sum_{i=1}^{d} \frac{n_i}{t_i - 1} \cdot \alpha^{i-1}$$
$$> \sum_{i=1}^{d} \frac{n_i}{t_i - 1} \cdot \frac{b_i}{c_i} = \sum_{i=1}^{d} \frac{HB}{t_i - 1}$$
$$\geqslant (h_\infty - \delta)\mathrm{OPT}(L), \tag{12}$$

where we used (8), (5) and (10). But now the inequality in (12) contradicts (4). □

## 4. A shelf algorithm with harmonic slave

In this section, we investigate the shelf algorithms SHELF$(H_k, \alpha)$ that use the *Harmonic* algorithms introduced by Lee and Lee [8] as slaves. Our analysis is strongly based on the techniques of [8].

The one-dimensional Harmonic bin packing algorithm $H_k$ behaves as follows. It is based on a nonuniform partition of the interval $[0,1]$ into $k$ parts $I_1, \ldots, I_k$ where $I_j = (1/(j+1), 1/j]$ for $1 \leqslant j \leqslant k-1$ and $I_k = [0, 1/k]$. To each of these intervals $I_j$, there always corresponds a single open bin, and only items belonging to $I_j$ are packed into this corresponding bin. If a new item arrives that does not fit into its corresponding bin, this bin is closed (and never opened again), and a new bin becomes the corresponding bin.

The function $g_k : [0,1] \to [0,1]$ is defined by

$$g_k(x) = \begin{cases} 1/j & \text{if } x \in I_j \text{ with } 1 \leqslant j < k, \\ kx/(k-1) & \text{if } x \in I_k. \end{cases} \tag{13}$$

Moreover, define the integer $u$ from $t_u \leqslant k < t_{u+1}$, and the number $G_k$,

$$G_k = \left(\sum_{l=1}^{u} \frac{1}{t_l - 1}\right) + \frac{k}{(t_{u+1} - 1)(k-1)}. \tag{14}$$

**Proposition 2** (Lee and Lee [8]).

(i) *For $k \geqslant 2$, $G_k > h_\infty$. As $k$ tends to infinity, $G_k$ tends to $h_\infty$.*

(ii) *For any integer $k \geqslant 2$, and for any sequence $x_1, \ldots x_m$ of real numbers, with $0 \leqslant x_i \leqslant 1$ and $\sum_{i=1}^{m} x_i \leqslant 1$, the inequality $\sum_{i=1}^{m} g_k(x_i) \leqslant G_k$ holds.* □

For $L$ an input list for the strip packing problem with rectangles $r_i = (w_i, h_i)$, $1 \leqslant i \leqslant n$, of width $w_i$ and height $h_i$, we define the *weight* of $r_i$ as $W(r_i) = h_i g_k(w_i)$, and the weight of the list $L$ as $W(L) = \sum_{i=1}^{n} W(r_i)$.

**Lemma 3.** *For any $k \geqslant 2$ and for any list $L$ over $[0,1]^2$, $W(L) \leqslant G_k \mathrm{OPT}(L)$ holds.*

**Proof.** Consider an arbitrary horizontal line that cuts through the optimum packing. This line cuts through several rectangles with widths, say, $w_{i_1}, \ldots, w_{i_m}$. All values $w_{i_j}$ are in $[0,1]$ and their sum is at most 1. Hence, by Proposition 2(ii), the total weight $g_k(\cdot)$ along this line is at most $G_k$. Now by integrating over the height of the optimal packing, the weight $W(L)$ of the list is seen to be at most $G_k$ times the height of the optimal packing. □

**Lemma 4.** *For any $k \geqslant 2$, for any $0 < \alpha < 1$, and for any list $L$ over $[0,1]^2$, the shelf algorithm $S =$ SHELF$(H_k, \alpha)$ yields a packing that satisfies $S(L) \leqslant W(L)/\alpha + k/(1-\alpha)$.*

**Proof.** For $s \geqslant 0$ and $1 \leqslant j \leqslant k$, denote by $L_{sj}$ the sublist of $L$ that contains the rectangles with width $w_i \in I_j = (1/(j+1), 1/j]$ and with height $\alpha^{s+1} < h_i \leqslant \alpha^s$.

Consider the packing produced by algorithm $S$ for list $L$. Clearly, all rectangles in $L_{sj}$ have been packed sequentially into shelves of height $\alpha^s$. While doing this, algorithm $S$ has filled and closed $z_{sj} \geqslant 0$ shelves, and there is at most one shelf of height $\alpha^s$ that remained open for items with width in $I_j$ after packing $L_{sj}$.

- If $1 \leqslant j < k$, then every closed shelf contains exactly $j$ items. Since the height of every item is at least $\alpha^{s+1}$ and its weight equals $1/j$, the total weight of every closed shelf is at least $j$ times $\alpha^{s+1}/j$. Thus $W(L_{sj}) \geqslant z_{sj}\alpha^{s+1}$.
- In case $j = k$, in every closed shelf the total width of the items is at least $1 - 1/k$: the shelf was closed since an item of width at most $1/k$ did not fit. Hence, the total weight of every closed shelf is at least $(1 - 1/k)$ times $\alpha^{s+1}k/(k-1)$. This yields $W(L_{sj}) \geqslant z_{sj}\alpha^{s+1}$.

Summarizing, for any $s$ and $j$, the total weight of all items in $L_{sj}$ is at least $z_{sj}\alpha^{s+1}$. On the other hand, $S$

uses a height of at most $(z_{sj}+1)\alpha^s$ for packing all items in $L_{sj}$ (where the "+1" one accounts for the single shelf that possibly remained open in the end). This gives

$$\begin{aligned} W(L) &= \sum_{j=1}^{k}\sum_{s=0}^{\infty} W(L_{sj}) \geqslant \sum_{j=1}^{k}\sum_{s=0}^{\infty} z_{sj}\alpha^{s+1} \\ &= \sum_{j=1}^{k}\sum_{s=0}^{\infty}(z_{sj}+1)\alpha^{s+1} - \sum_{j=1}^{k}\sum_{s=0}^{\infty}\alpha^{s+1} \\ &\geqslant \alpha S(L) - \frac{k\alpha}{1-\alpha}. \end{aligned} \tag{15}$$

Dividing this inequality by $\alpha$ completes the proof of the lemma. □

**Theorem 5.** *For any $\delta > 0$, there exists a $k \geqslant 2$ and an $0 < \alpha < 1$ such that the shelf algorithm $S =$ SHELF$(H_k, \alpha)$ has asymptotic worst case ratio $R_S^\infty \leqslant h_\infty + \delta$.*

**Proof.** For every $k \geqslant 2$ and for every $\alpha$, $0 < \alpha < 1$, we may combine the statements in Lemmas 3 and 4 to get

$$\begin{aligned} S(L) &\leqslant \frac{W(L)}{\alpha} + \frac{k}{1-\alpha} \\ &\leqslant \frac{G_k}{\alpha}\mathrm{OPT}(L) + \frac{k}{1-\alpha}. \end{aligned} \tag{16}$$

Hence, the asymptotic worst case ratio of the corresponding shelf algorithm is at most $G_k/\alpha$. Because of Proposition 2(i), by choosing $k$ sufficiently large and $\alpha$ sufficiently close to 1, we can bring the asymptotic worst case ratio arbitrarily close to $h_\infty$. □

## 5. Discussion

At the end of this note we know that there exists a shelf algorithm for on-line strip packing with asymptotic worst case ratio arbitrarily close to 1.69103 and we also know that no shelf algorithm can improve on this. The question arises of finding better on-line algorithms that are not based on the shelf concept. How close can we come to the 1.5401 lower bound for general on-line strip packing algorithms?

Whereas the FIRST FIT (with asymptotic worst case ratio 1.7) and the Harmonic algorithm (with asymptotic worst case ratio 1.69103) for the one-dimensional on-line bin packing problem lead to on-line strip packing algorithms with similar asymptotic worst case ratios, the situation is less clear for other one-dimensional on-line bin packing algorithms: Richey's algorithm [9] has an asymptotic worst case ratio of less than 1.5888. Can this yield a strip packing algorithm of similar quality?

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